

# Cognitive Load in AI-Assisted Systems

Three self-contained intern tracks derived from the full research proposal. Each track has a defined scope, step-by-step implementation guide, intermediate outputs the team can review, and by-products that feed other tracks or the broader project.

| # | Track                            | Core task                                                                        | Primary output                                                         |
|---|----------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 1 | Interaction Logger & Task Lab    | Build AI task scenarios, log interaction events, collect NASA-TLX ratings        | Event-annotated interaction log + subjective ratings dataset           |
| 2 | Signal Collection & CLS Pipeline | Set up EEG / HRV / eye-tracker data collection and build the CLS fusion pipeline | Synchronized multimodal dataset + Composite CLS Score per event        |
| 3 | ML Prediction & UX Audit         | Train a CLS predictor on behavioral data; audit interfaces for overload hotspots | Deployed CLS prediction model + UX overload report + design guidelines |

## Track 1 — Interaction Logger & Task Lab

**Problem statement:** AI tools are widely used but the exact cognitive events they trigger — a confusing suggestion, a trust failure, a decision overload moment — are never recorded. Without a structured log of what happened during an AI interaction and how mentally effortful it felt, there is no data to measure. This track builds that foundation: a browser-based interaction logger wired to structured AI tasks, yielding the event dataset that all other tracks depend on.

|                   |                                                                                       |
|-------------------|---------------------------------------------------------------------------------------|
| <b>Duration</b>   | 3–4 weeks                                                                             |
| <b>Team size</b>  | 1–2 interns (HCI / web dev background preferred)                                      |
| <b>Tools</b>      | Python · JavaScript · Flask or FastAPI · NASA-TLX JS widget · JSON / SQLite           |
| <b>Feeds into</b> | Track 2 (event timestamps for synchronization) · Track 3 (behavioral features for ML) |

### Implementation steps

#### Define task scenarios

- 1 Design 6–8 structured AI interaction tasks spanning content generation (summarization, drafting), decision support (choosing between AI options), and information retrieval. Each task must have a controlled complexity variant (simple / medium / complex) and a fixed time window. Write a task protocol document.

#### Build the interaction logger

- 2 Develop a lightweight browser extension or web wrapper that captures: prompt submitted, AI response received, user edits made, response accepted/rejected, time between each event (latency), scroll depth, and copy-paste events. Log all events as JSON with ISO timestamps to a local SQLite database.

#### Embed NASA-TLX and Paas scale

- 3 After each task, automatically surface a digital NASA-TLX form (6 dimensions: mental demand, physical demand, temporal demand, performance, effort, frustration) and the single-item Paas cognitive load scale. Store ratings linked to task ID and session ID.

#### Run pilot sessions ( $N \geq 30$ )

- 4 Recruit participants across at least two user groups (students, working professionals). Run each participant through all task scenarios in counterbalanced order. Collect demographic data (age, AI familiarity, domain). Target at least 5 sessions per task-complexity combination.

#### Validate and clean the log

- 5 Write a data quality script: flag sessions with missing NASA-TLX, implausibly short latencies ( $< 2s$ ), or incomplete task completion. Compute inter-rater agreement on task difficulty ratings. Export a clean, de-identified CSV with one row per interaction event.

- 6 **Produce event-level annotation**

Manually annotate a 20% sample of logs to label high-load events (defined as: latency > 2× median for that task + NASA-TLX > 60 + edit rate > 3 edits/response). This becomes the ground-truth seed for Track 3's ML model.

### **By-products**

- **Open-source AI interaction logger:** reusable by any HCI lab without this project's infrastructure
- **Normative NASA-TLX baselines for generative AI tasks:** first such dataset for Indian user populations
- **Task complexity taxonomy:** a reusable classification of AI task types by expected cognitive demand
- **Participant panel:** 30+ consented participants available for Track 2 physiological sessions

## Track 2 — Signal Collection & CLS Pipeline

**Problem statement:** Subjective ratings tell you how someone *felt* about a task *after* it ended. Physiological signals tell you what their nervous system was *doing* during it — *but only if those* signals are cleanly collected, time-aligned with specific interaction events, and converted into a single interpretable score. This track solves the synchronization and fusion problem: connecting EEG, HRV, and eye-tracking streams to the event log from Track 1, and computing a Composite Cognitive Load Score (CCLS) that is anchored to each AI interaction event.

|            |                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------|
| Duration   | 4–5 weeks (weeks 2–6, overlaps with Track 1 from week 2)                                                                      |
| Team size  | 1–2 interns (signal processing / bioinformatics / Python background preferred)                                                |
| Tools      | Python · MNE-Python (EEG) · neurokit2 (HRV) · Tobii SDK or PyGaze (eye tracking) · LSL (Lab Streaming Layer) · pandas · scipy |
| Depends on | Track 1 event log and timestamps (needed for synchronization)                                                                 |
| Feeds into | Track 3 (CCLS labels as ML training targets)                                                                                  |

### Implementation steps

#### Set up sensor hardware and LSL streams

- 1 Configure EEG (2 units), HRV wristband, and eye tracker to stream data via Lab Streaming Layer (LSL). Write an LSL recorder script that captures all streams simultaneously into a single XDF file per session. Validate latency: all streams must be within  $\pm 50$  ms of each other.

#### Synchronize with interaction log

- 2 Inject LSL markers at each interaction event (prompt submitted, response received, edit made, NASA-TLX shown) from the Track 1 logger. Write a sync script that aligns XDF physiological streams to JSON interaction events using shared epoch timestamps. Output: one merged DataFrame per session with physio + event columns.

#### Preprocess EEG signal

- 3 Apply bandpass filter (0.5–45 Hz), re-reference to average mastoids, run ICA to remove eye blink and muscle artifacts using MNE-Python. Extract per-event epochs (–0.5 s to +3 s around each event). Compute power spectral density in theta (4–8 Hz) and alpha (8–13 Hz) bands — both are validated EEG markers of cognitive workload.

#### Preprocess HRV and eye tracking

- 4 For HRV: detect R-peaks, compute SDNN, RMSSD, and LF/HF ratio per 30-second window using neurokit2. For eye tracking: compute mean fixation duration, blink rate, and pupil diameter per interaction event window. Flag and exclude windows with > 20% artifact contamination.

#### Compute Composite Cognitive Load Score (CCLS)

- 5 Normalize each signal to z-scores within participant. Compute CCLS as a weighted sum:  $CCLS = 0.40 \times NASA\_TLX\_z + 0.25 \times EEG\_theta\_z + 0.20 \times HRV\_LF/HF\_z + 0.15 \times pupil\_diam\_z$ . Initial

weights from literature (Kosch et al. 2023), refine via confirmatory factor analysis after pilot run.  
Output: one CCLS value per interaction event.

### **Validate CCLS against task complexity**

- 6 Run a mixed-effects model:  $CCLS \sim task\_complexity + task\_type + participant\_group + (1|participant)$ . Expect CCLS to increase monotonically with task complexity. Compute Pearson correlation between CCLS and NASA-TLX total score (target  $r > 0.6$ ). Document any signals that do not correlate.

### **By-products**

- **Reusable LSL synchronization toolkit:** open-source Python package for aligning physio streams to browser interaction logs
- **Preprocessed EEG / HRV / eye feature library:** reusable signal features across future cognitive studies at C-DAC
- **CCLS formula with empirically validated weights:** methodology contribution publishable independently
- **Normative physiological baselines for AI interaction tasks:** first such reference values for Indian adult populations
- **Lab SOP (standard operating procedure):** for multimodal cognitive load data collection — reusable for future CDAC studies

## Track 3 — ML Prediction & UX Audit

**Problem statement:** Physiological sensors cannot be attached to every user of an AI system. For cognitive load measurement to be practically useful — to inform real-time adaptive AI, product design decisions, and policy guidelines — it must work from behavioral data alone: what the user typed, how long they hesitated, how many times they edited a response. This track trains a machine learning model to predict CCLS from the behavioral event log produced by Tracks 1 and 2, and uses that model to audit real AI interfaces for cognitive overload hotspots.

|                   |                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------|
| <b>Duration</b>   | 2 weeks (starts week 3, after Track 2 produces CCLS labels)                                              |
| <b>Team size</b>  | 1–2 interns (ML / data science background required)                                                      |
| <b>Tools</b>      | Python · scikit-learn · XGBoost · SHAP · pandas · matplotlib · seaborn · Streamlit (for audit dashboard) |
| <b>Depends on</b> | Track 1 behavioral CSV · Track 2 CCLS labels (ML training targets)                                       |
| <b>Produces</b>   | Deployable CLS prediction model · UX overload report · Design guidelines                                 |

### Implementation steps

#### Feature engineering from interaction log

- 1 From the Track 1 event CSV, engineer behavioral features per interaction event: response latency (ms), edit count, edit-to-accept ratio, prompt revision count, word count of prompt vs response, scroll depth, time-on-page, session duration, task sequence position, number of AI suggestions evaluated. Produce a feature matrix with one row per event.

#### Merge with CCLS labels

- 2 Join the behavioral feature matrix with Track 2's CCLS scores on session\_id + event\_id. Handle missing CCLS values (sessions where physio was not collected) with listwise deletion. Result: a labeled dataset of ~N×8 features + CCLS target. Bin CCLS into three classes (low / medium / high) for classification experiments alongside regression.

#### Train and compare models

- 3 Train and cross-validate (5-fold, stratified by participant): Linear Regression baseline, Random Forest, XGBoost, and a lightweight MLP. Evaluate on RMSE (regression) and F1 macro (classification). Use SMOTE if class imbalance > 2:1. Record all results in a model comparison table. Target: F1 macro > 0.70 for the classification task.

#### Explainability with SHAP

- 4 Apply SHAP on the best model. Produce: (a) global feature importance bar chart — which behavioral signals most predict high CCLS; (b) per-event SHAP waterfall plots showing why a specific event scored high. This analysis directly informs the UX audit in Step 5.

#### UX audit of three AI interfaces

- 5 Select three real AI-assisted interfaces (e.g. ChatGPT, a government AI chatbot, an AI tutoring tool). Run 10 participants through the Track 1 task protocol on each interface. Apply the trained CLS predictor to the interaction logs. Identify UX Overload Points: interface states where

predicted CCLS exceeds the 75th percentile. Map overload points to specific UI elements (suggestion cards, confidence scores, alert banners, response length).

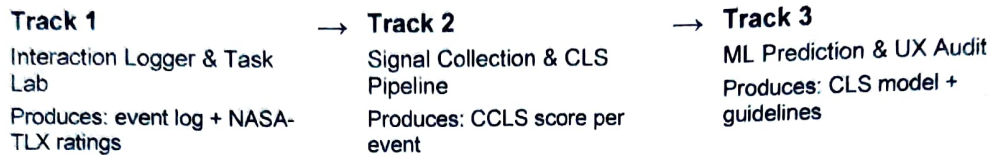
### **Write design guidelines and audit report**

- 6 Synthesize SHAP findings and UX Overload Points into a structured design guideline document: 8–12 concrete, evidence-backed rules for reducing cognitive load in AI interfaces (e.g. 'limit simultaneous AI suggestions to  $\leq 3$ '; 'provide rationale with high-stakes recommendations'; 'allow user-controlled response verbosity'). Produce a one-page overload point map per audited interface.

### **By-products**

- **Behavioral feature engineering library:** open Python package for extracting cognitive load features from any browser interaction log
- **CLS model card:** standardized documentation of model performance, limitations, and intended use — following Responsible AI best practices
- **Comparative interface audit:** the first published cognitive load benchmarking of major AI chat interfaces used in India
- **Evidence base for policy:** quantitative overload measurements that MeitY, NITI Aayog, and ABDM can cite in AI interface procurement standards
- **Streamlit audit dashboard:** a tool any non-technical team member can use to run the CLS predictor on a new interaction log and visualise overload hotspots

## How the three tracks connect



Tracks 1 and 2 can run in parallel from week 2 once the task protocol and logger are stable. Track 3 begins as soon as Track 2 produces its first CCLS-labeled batch (week 3). All three tracks share participants, the task protocol, and the interaction log schema — changes to any of these must be version-controlled and communicated across all active teams.

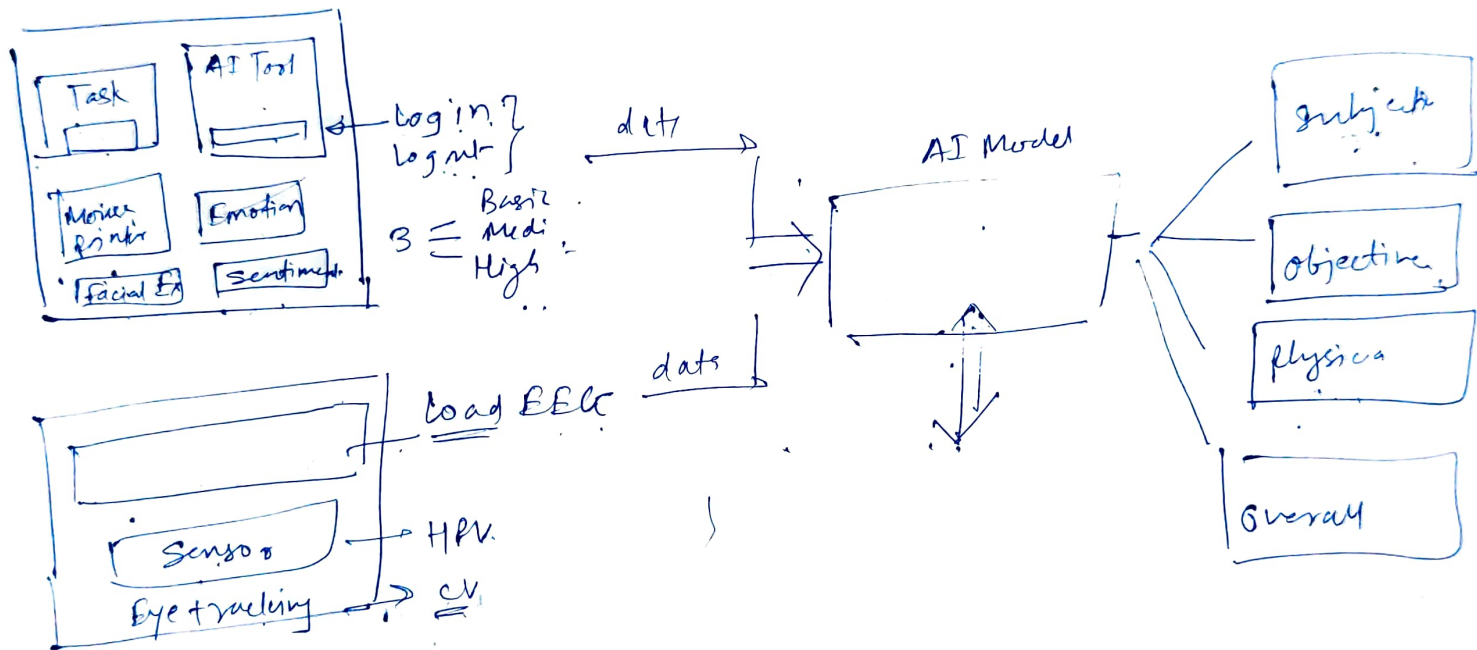
**Combined output of all three tracks:** Event-anchored annotated dataset · Validated CCLS formula · Deployable ML prediction model · UX overload maps · Evidence-based design guidelines — together forming the full Cognitive Load Measurement Framework (CLMF) the proposal commits to deliver.

AI Tool ⇒ switch, +ve sentiment

Task ⇒  Task related with

Eye Tracking, Mouse Tracking, Emotion

subjective  
objective  
physiological } Multimodal dataset



NASD-TLx - 0 to 100